

Основные характеристики датасета

Датасет: Online Retail II

Размер: 541910 строк

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/10 8:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	12/1/10 8:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/10 8:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/10 8:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/10 8:26	3.39	17850.0	United Kingdom
5	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	12/1/10 8:26	7.65	17850.0	United Kingdom
6	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	12/1/10 8:26	4.25	17850.0	United Kingdom
7	536366	22633	HAND WARMER UNION JACK	6	12/1/10 8:28	1.85	17850.0	United Kingdom
8	536366	22632	HAND WARMER RED POLKA DOT	6	12/1/10 8:28	1.85	17850.0	United Kingdom
9	536368	22960	JAM MAKING SET WITH JARS	6	12/1/10 8:34	4.25	13047.0	United Kingdom

```

Column types:
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RangeIndex: 541910 entries, 0 to 541909
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
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0   Invoice          541910 non-null object
1   StockCode        541910 non-null object
2   Description      540456 non-null object
3   Quantity         541910 non-null int64
4   InvoiceDate       541910 non-null object
5   Price            541910 non-null float64
6   Customer ID      406830 non-null float64
7   Country          541910 non-null object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB

```

	Quantity	Price	Customer ID
count	541910.000000	541910.000000	406830.000000
mean	9.552234	4.611138	15287.684160
std	218.080957	96.759765	1713.603074
min	-80995.000000	-11062.060000	12346.000000
25%	1.000000	1.250000	13953.000000
50%	3.000000	2.080000	15152.000000
75%	10.000000	4.130000	16791.000000
max	80995.000000	38970.000000	18287.000000

1. Пропуски присутствуют в столбцах Description и Customer ID.
2. Для анализа покупателей создадим датасет `df_customers`, удалив строки без Customer ID. Пропуски в Description заменим на "Unknown".
3. Для анализа товаров и продаж создадим датасет `df_sales`, сохранив все строки, включая покупки без Customer ID. Пропуски в Description также заменим на "Unknown".

Удаление выбросов

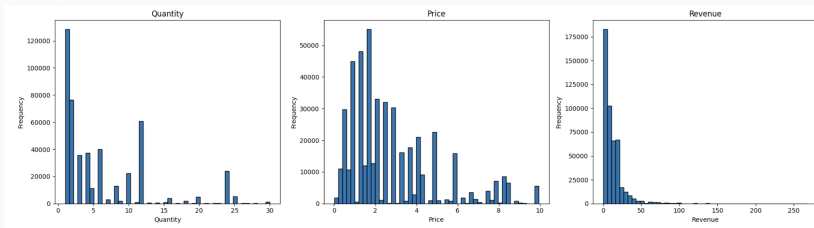
- Quantity 0.95 quantile: 30.0
- Price 0.95 quantile: 9.95
- Rows with Quantity > 30.0: 25603
- Rows with Price > 9.95: 24476

Invoice	StockCode	Description	Quantity	Price	Country
581483	23843	PAPER CRAFT , LITTLE BIRDIE	80995	2.08	United Kingdom
541431	23166	MEDIUM CERAMIC TOP STORAGE JAR	74215	1.04	United Kingdom
573808	84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS	4800	0.21	United Kingdom
554868	22197	SMALL POPCORN HOLDER	4300	0.72	United Kingdom
544612	22053	EMPIRE DESIGN ROSETTE	3906	0.82	United Kingdom
560599	18007	ESSENTIAL BALM 3.5g TIN IN ENVELOPE	3186	0.06	United Kingdom
550461	21108	FAIRY CAKE FLANNEL ASSORTED COLOUR	3114	2.10	United Kingdom
540815	21108	FAIRY CAKE FLANNEL ASSORTED COLOUR	3114	2.10	United Kingdom
573995	16014	SMALL CHINESE STYLE SCISSOR	3000	0.32	United Kingdom
536830	84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS	2880	0.18	United Kingdom
562439	84879	ASSORTED COLOUR BIRD ORNAMENT	2880	1.45	United Kingdom
554272	21977	PACK OF 60 PINK PAISLEY CAKE CASES	2700	0.42	United Kingdom
543057	84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS	2592	0.21	United Kingdom
544152	18007	ESSENTIAL BALM 3.5g TIN IN ENVELOPE	2400	0.06	United Kingdom
539101	22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2400	0.94	United Kingdom
543669	22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2400	0.94	United Kingdom
573003	23084	RABBIT NIGHT LIGHT	2400	2.08	Netherlands
547037	21957	PACK OF 12 SKULL TISSUES	2160	0.25	United Kingdom
574294	21915	RED HARMONICA IN BOX	2100	1.06	United Kingdom
579498	23084	RABBIT NIGHT LIGHT	2040	1.79	Japan
Invoice	StockCode	Description	Quantity	Price	Country
537632	AMAZONFEE	AMAZON FEE	1	13541.33	United Kingdom
A563185	B	Adjust bad debt	1	11062.06	United Kingdom
551697	POST	POSTAGE	1	8142.75	United Kingdom
562955	DOT	DOTCOM POSTAGE	1	4505.17	United Kingdom
560373	M	Manual	1	4287.63	United Kingdom
573880	M	Manual	1	4161.06	France
573077	M	Manual	1	4161.06	France
571751	M	Manual	1	3949.32	Singapore
569382	M	Manual	1	3155.95	United Kingdom
567353	M	Manual	1	2653.95	Hong Kong
546558	M	Manual	1	2583.76	Hong Kong
562946	M	Manual	1	2500.00	United Kingdom
548813	M	Manual	1	2382.92	Singapore
578149	DOT	DOTCOM POSTAGE	1	2275.54	United Kingdom
580610	DOT	DOTCOM POSTAGE	1	2196.67	United Kingdom
571751	M	Manual	1	2118.74	Singapore
580612	DOT	DOTCOM POSTAGE	1	2114.00	United Kingdom
548820	M	Manual	1	2053.07	Singapore
566927	M	Manual	1	2033.10	United Kingdom
578833	DOT	DOTCOM POSTAGE	1	2028.25	United Kingdom

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country	Revenue
157195	550193	PADS	PADS TO MATCH ALL CUSHIONS	1	2011-04-15 09:27:00	0.001	13952.0	United Kingdom	0.001
279045	561226	PADS	PADS TO MATCH ALL CUSHIONS	1	2011-07-26 10:13:00	0.001	15618.0	United Kingdom	0.001
359871	568200	PADS	PADS TO MATCH ALL CUSHIONS	1	2011-09-25 14:58:00	0.001	16198.0	United Kingdom	0.001
361741	568375	BANK CHARGES	Bank Charges	1	2011-09-26 17:01:00	0.001	13405.0	United Kingdom	0.001
217066	555876	16045	POPART WOODEN PENCILS ASST	100	2011-06-07 15:28:00	0.040	15799.0	United Kingdom	4.000
221726	556267	16045	POPART WOODEN PENCILS ASST	500	2011-06-09 19:33:00	0.040	13694.0	United Kingdom	20.000
228959	557006	16045	POPART WOODEN PENCILS ASST	100	2011-06-16 11:29:00	0.040	14282.0	United Kingdom	4.000
229052	557009	16045	POPART WOODEN PENCILS ASST	100	2011-06-16 11:38:00	0.040	14321.0	United Kingdom	4.000
229498	557057	16045	POPART WOODEN PENCILS ASST	100	2011-06-16 14:46:00	0.040	16843.0	United Kingdom	4.000
233804	557492	16045	POPART WOODEN PENCILS ASST	100	2011-06-20 15:08:00	0.040	13717.0	United Kingdom	4.000

	Quantity		InvoiceDate	Price	Customer ID	Revenue
count	474986.000000		474986	474986.000000	357590.000000	474986.000000
mean	6.153722	2011-07-05 01:55:14.378697472		2.779845	15314.494723	12.187144
min	1.000000	2010-12-01 08:26:00		0.060000	12347.000000	0.060000
25%	1.000000	2011-03-28 12:22:00		1.250000	13985.000000	3.360000
50%	3.000000	2011-07-20 14:55:00		2.080000	15214.000000	8.290000
75%	10.000000	2011-10-19 13:58:00		3.750000	16813.000000	16.500000
max	30.000000	2011-12-09 12:50:00		9.950000	18287.000000	268.650000
std	6.477024		NaN	2.189721	1709.482104	14.239283

Распределение признаков Quantity, Price, Revenue



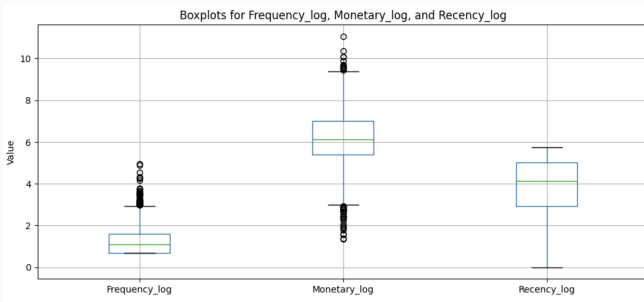
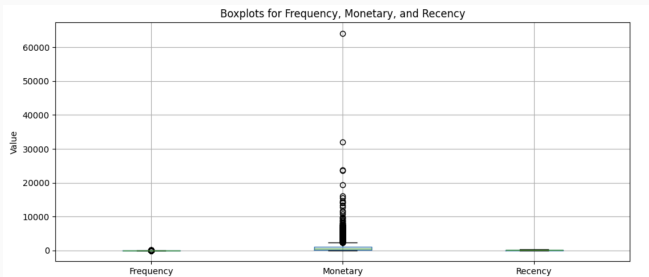
- Большинство покупок содержит небольшое количество товаров. Чаще всего покупают 1-5 единиц товара. Есть пики в точках 12 и 24: это похоже на покупку товаров упаковками или наборами по 12 и 24 штуки.
- Цены товаров в основном невысокие. Большинство товаров стоит от 1 до 5 денежных единиц.
- Выручка по одной строке заказа обычно невелика, для большинства позиций она составляет менее 20 денежных единиц.

Создание новых признаков

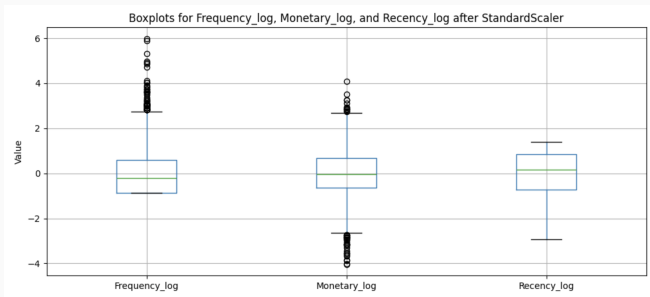
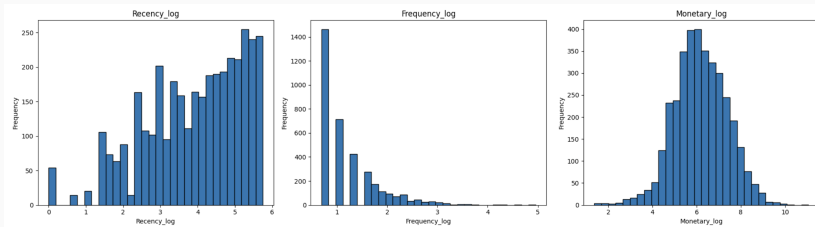
	Customer ID	Frequency	Monetary	Recency
0	12347	5	2262.71	69
1	12348	3	90.20	188
2	12350	1	294.40	249
3	12352	6	1090.26	11
4	12353	1	89.00	143
5	12354	1	840.09	171
6	12355	1	238.90	153
7	12356	2	1290.44	185
8	12358	1	404.86	90
9	12359	3	2306.93	129

1. **Frequency** - количество совершенных покупок
2. **Monetary** - потраченная сумма
3. **Recency** - количество дней, прошедших с момента последней покупки

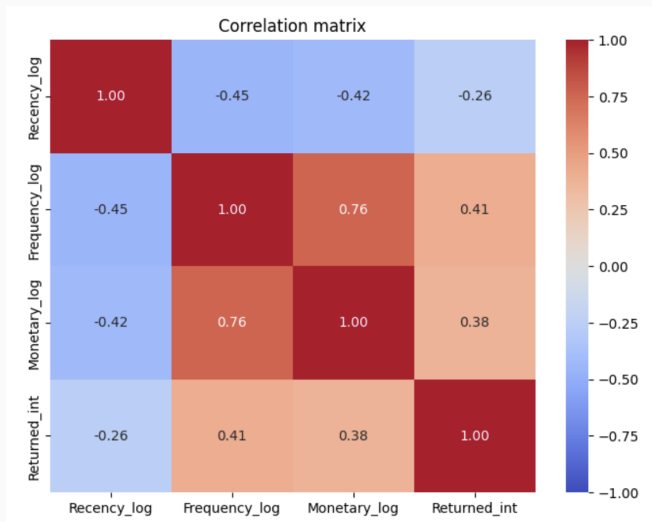
Логарифмирование признаков



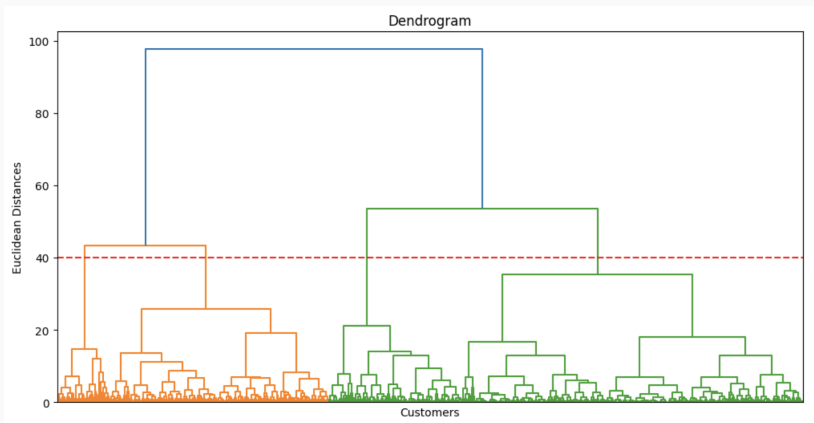
Распределение признаков Frequency, Monetary, Recency, нормализация



Создание целевой переменной, матрица корреляций

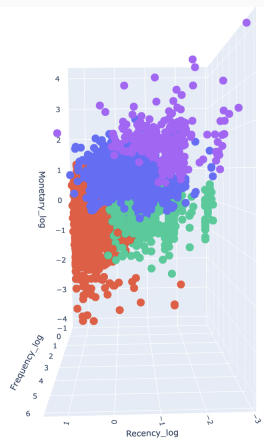
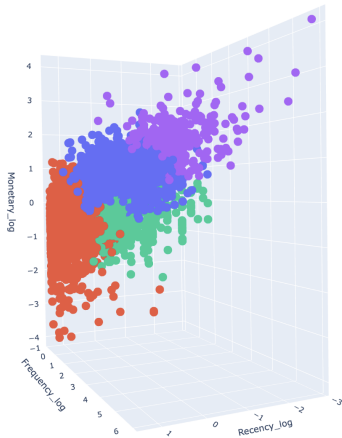


Иерархическая кластеризация



cluster	Recency_log	Frequency_log	Monetary_log	Count
1	0.816555	-0.617785	-0.628036	1644
2	-0.375991	0.708723	0.767548	1056
3	-0.914544	-0.513108	-0.358567	648
4	-1.361948	2.315527	1.754098	259

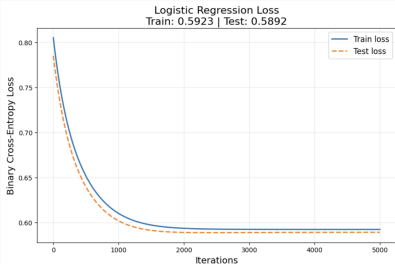
1. **Inactive:** давно не совершали покупок, редко и мало покупали.
2. **Loyal:** возвращались недавно, посещали магазин и тратили средства умеренно
3. **New:** небольшой временной промежуток с последней покупки, но редко посещали магазин и потратили меньше средств
4. **VIP:** совершали покупки недавно, при этом посещали магазин часто и тратили большое количество средств



segment

- Loyal
- Inactive
- New
- VIP

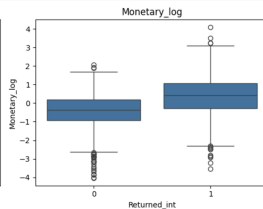
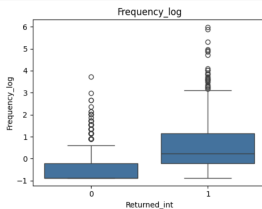
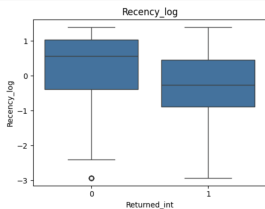
Линейная регрессия



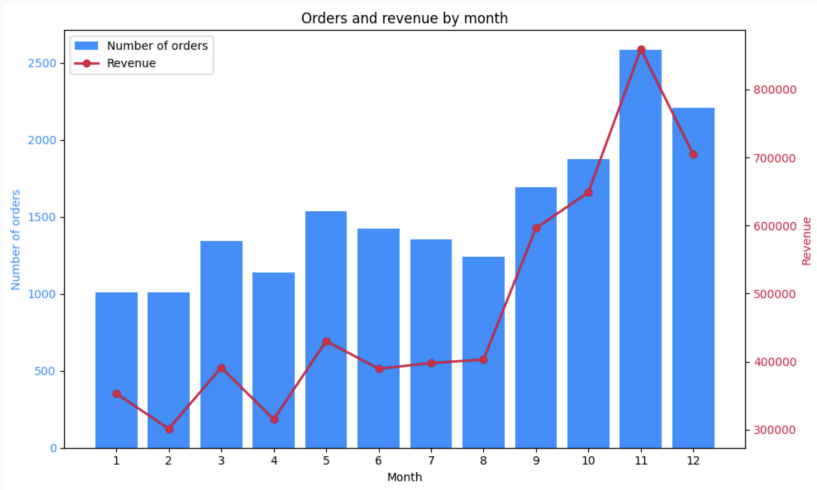
Train metrics				
	precision	recall	f1-score	support
0	0.68	0.78	0.72	1131
1	0.71	0.60	0.65	1033
accuracy			0.69	2164
macro avg	0.69	0.69	0.69	2164
weighted avg	0.69	0.69	0.69	2164
Test metrics				
	precision	recall	f1-score	support
0	0.68	0.74	0.71	736
1	0.70	0.63	0.66	707
accuracy			0.69	1443
macro avg	0.69	0.68	0.68	1443
weighted avg	0.69	0.69	0.69	1443

Train accuracy: 0.6913123844731978

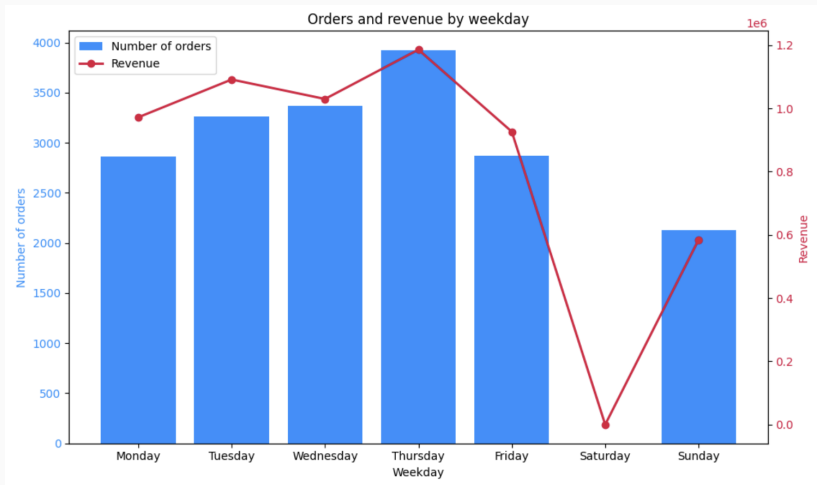
Test accuracy: 0.6860706860706861



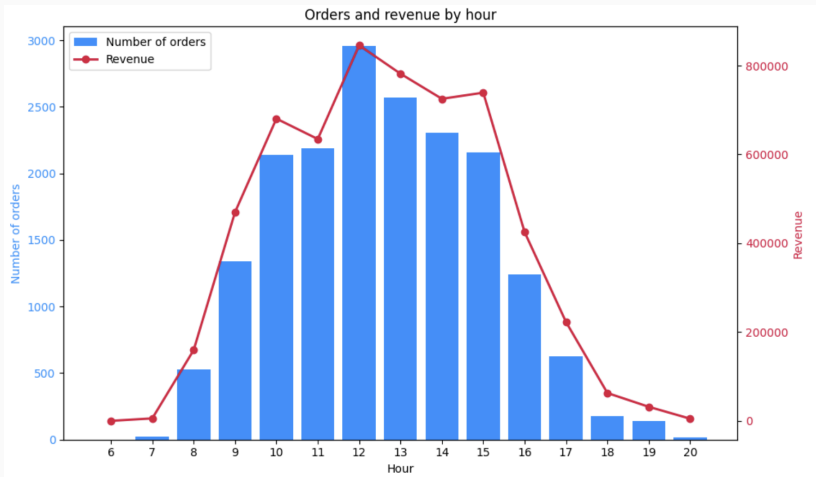
Анализ продаж по месяцам



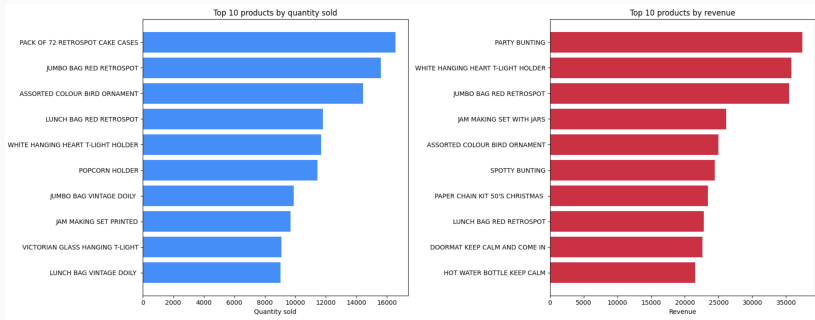
Анализ продаж по дням недели

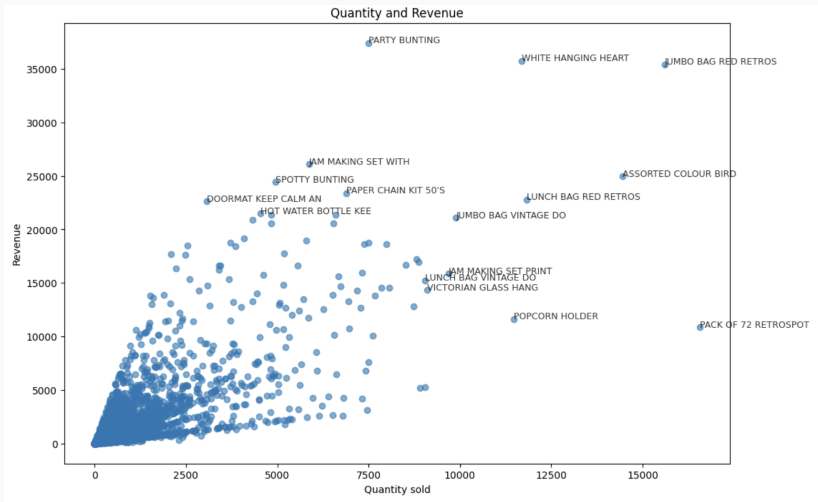


Анализ продаж по часам



Самые популярные и прибыльные товары



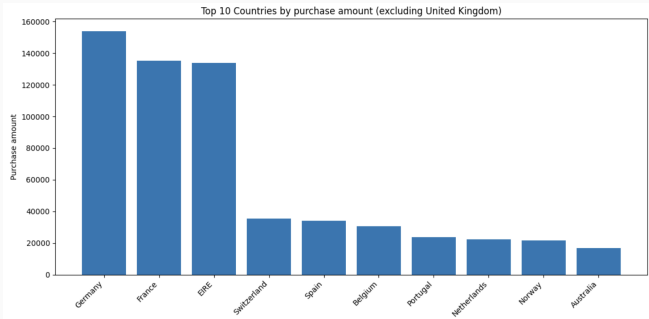


Популярные пары товаров

StockCode 1	Product 1	StockCode 2	Product 2	Count
22386	JUMBO BAG PINK POLKADOT	85099B	JUMBO BAG RED RETROSPOT	751
22697	GREEN REGENCY TEACUP AND SAUCER	22699	ROSES REGENCY TEACUP AND SAUCER	743
21931	JUMBO STORAGE BAG SUKI	85099B	JUMBO BAG RED RETROSPOT	669
22411	JUMBO SHOPPER VINTAGE RED PAISLEY	85099B	JUMBO BAG RED RETROSPOT	633
22726	ALARM CLOCK BAKELIKE GREEN	22727	ALARM CLOCK BAKELIKE RED	624
20725	LUNCH BAG RED RETROSPOT	22383	LUNCH BAG SUKI DESIGN	622
22697	GREEN REGENCY TEACUP AND SAUCER	22698	PINK REGENCY TEACUP AND SAUCER	614
20725	LUNCH BAG RED RETROSPOT	20727	LUNCH BAG BLACK SKULL.	610
22698	PINK REGENCY TEACUP AND SAUCER	22699	ROSES REGENCY TEACUP AND SAUCER	577
20725	LUNCH BAG RED RETROSPOT	22384	LUNCH BAG PINK POLKADOT	570
20727	LUNCH BAG BLACK SKULL.	22383	LUNCH BAG SUKI DESIGN	567
20725	LUNCH BAG RED RETROSPOT	22382	LUNCH BAG SPACEBOY DESIGN	534
85099B	JUMBO BAG RED RETROSPOT	85099C	JUMBO BAG BAROQUE BLACK WHITE	534
23203	JUMBO BAG VINTAGE DOILY	85099B	JUMBO BAG RED RETROSPOT	534
20725	LUNCH BAG RED RETROSPOT	85099B	JUMBO BAG RED RETROSPOT	532
20727	LUNCH BAG BLACK SKULL.	22384	LUNCH BAG PINK POLKADOT	528
20725	LUNCH BAG RED RETROSPOT	20728	LUNCH BAG CARS BLUE	522
20728	LUNCH BAG CARS BLUE	22383	LUNCH BAG SUKI DESIGN	521
22382	LUNCH BAG SPACEBOY DESIGN	22383	LUNCH BAG SUKI DESIGN	520
23300	GARDENERS KNEELING PAD CUP OF TEA	23301	GARDENERS KNEELING PAD KEEP CALM	512

Анализ наиболее частых товарных пар показал, что покупатели часто приобретают вместе товары одной серии, одного типа или схожего дизайна. Особенно заметны пары среди сумок, чайных наборов и декоративных товаров.

Анализ продаж по странам

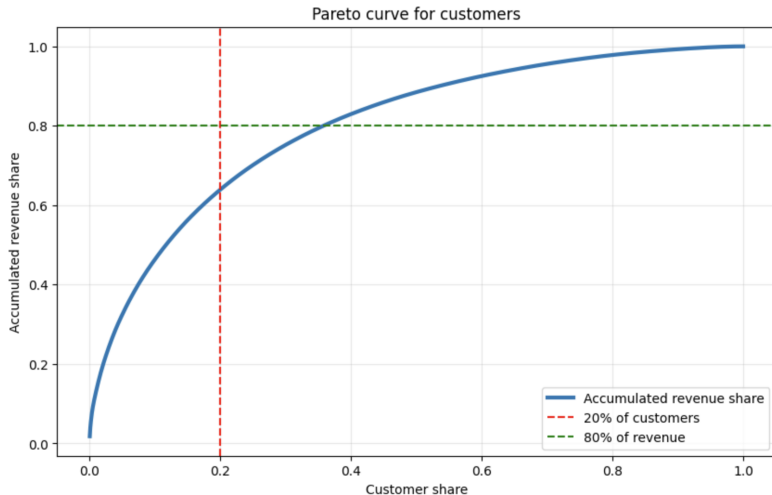


Number of different countries: 38

	Country	Revenue
1	United Kingdom	5058933.83
2	Germany	154037.38
3	France	135268.52
4	EIRE	133930.70
5	Switzerland	35471.48
6	Spain	33904.39
7	Belgium	30706.48
8	Portugal	23829.01
9	Netherlands	22396.95
10	Norway	21723.36

Кривая Парето

The top 20% of customers generate 63.9% of revenue





Спасибо за внимание!